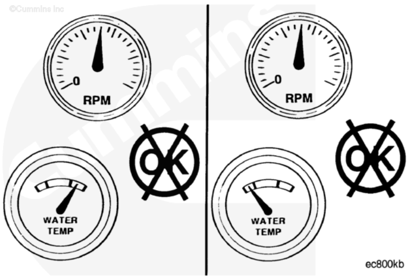
Trainings

Initial Check

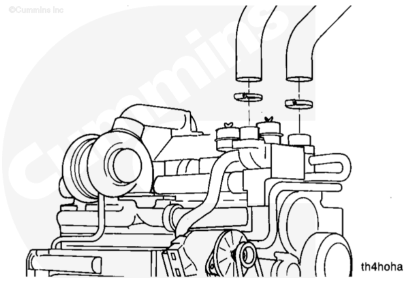
All Applications

The engine thermostat and thermostat seal **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.



⚠ WARNING ⚠

Complete this test with the engine coolant temperature below 50°C [120°F]. Hot steam can cause serious personal injury.



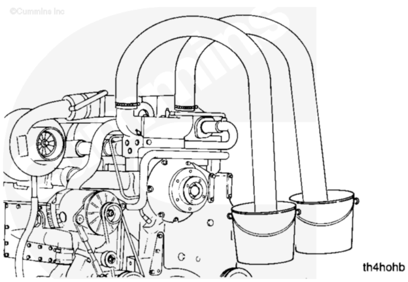
If the engine coolant temperature is below normal, perform the steps below.

Remove the upper engine radiator hoses from the thermostat housing.

Install two hoses of the same size on the thermostat housing outlets long enough to reach a remote dry container.

Install and tighten hose clamps on the housing outlets.

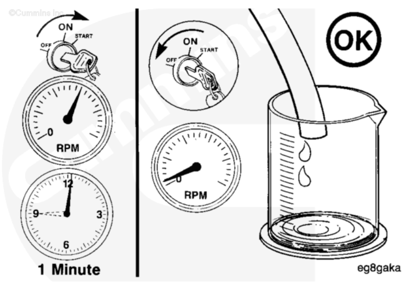
Place the end of the hoses in two dry containers.



Operate the engine at rated rpm for 1 minute.

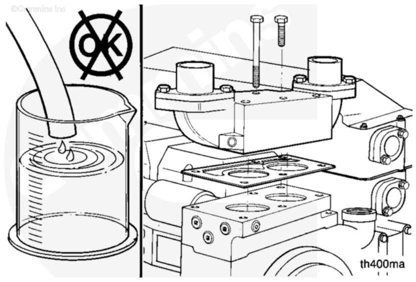
Shut the engine off, and measure the amount of coolant collected in each container.

The amount of coolant collected **must not** be more than 100 cc [3.3 fl oz].



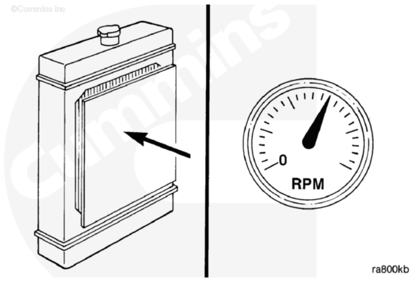
If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat or the thermostat seal is leaking.

Remove and test the thermostat.



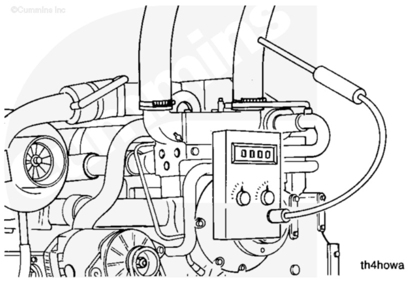
If the engine coolant temperature is above normal, perform the steps below

Restrict the radiator air flow. Operate the engine until the coolant temperature rises to 90 to 93°C [195 to 200°F].

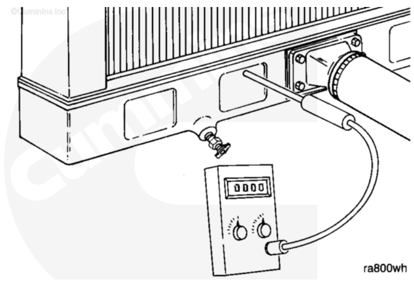


Record the temperature of the coolant outlet hoses. An increase in temperature indicates the thermostats have started to open.

Use a contact pyrometer (shown) or install a temperature gauge in the desired locations prior to operating the engine.



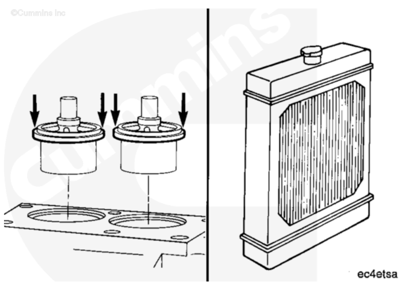
Record the temperature of the radiator bottom tank or coolant out tube.



If the difference in temperature is more than 8°C [15°F], either the thermostats are **not** fully open or the radiator core is plugged.

Replace the thermostats.

To check the radiator, Refer to Procedure 008-042 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-042.html)

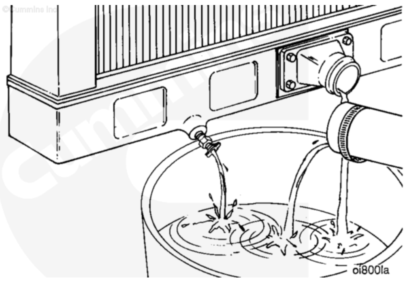


Preparatory Steps

All Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

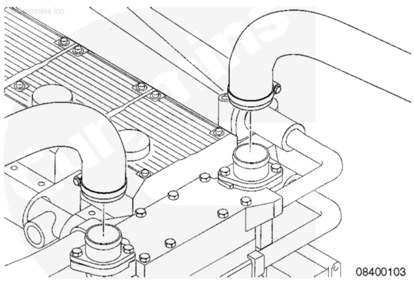
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html)

Remove

Industrial Applications

Remove both upper radiator hoses from the thermostat housing.

Remove the vent lines from the top of the thermostat housing.

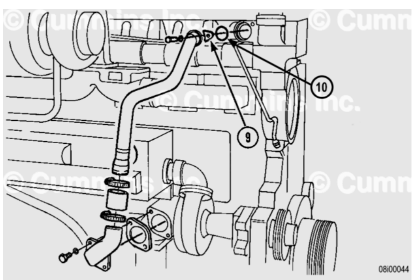


If the engine is equipped with an air compressor, remove the air compressor coolant return tube.

Remove bypass tube clip (9).

Remove and discard the seal (10).

Loosen both hose clamps.

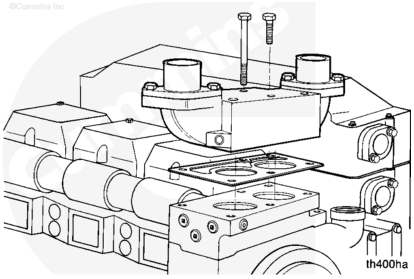


Remove the bypass tube.

Remove the eight long capscrews.

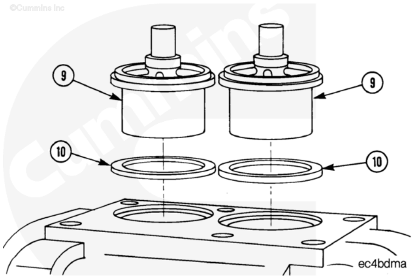
Remove the thermostat housing.

Remove and discard the gasket.



Remove the thermostats (9).

Remove the seals (10). Refer to Procedure 008-016 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-016.html)



Marine Applications

Remove the three captive washer capscrews from the water transfer connection.

Pull the water transfer connection away from the thermostat housing.

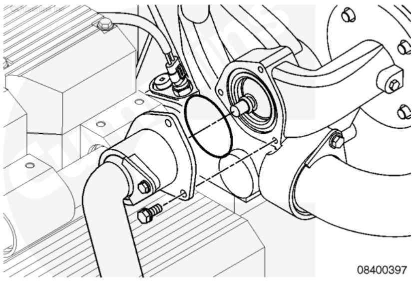
The pipe will rotate around the lower end.

Discard the o-ring.

Remove the thermostat.

Remove the thermostat seal.

Refer to Procedure 008-016 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-016.html)

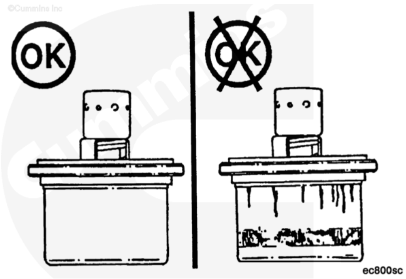


Inspect for Reuse

All Applications

Inspect the thermostat for damage.

If the thermostat is damaged, it **must** be replaced.



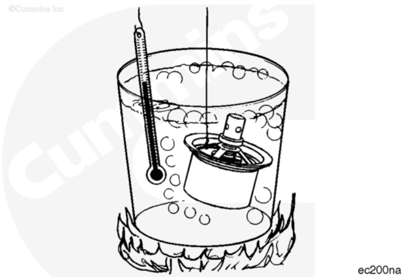
Test

Industrial Applications

Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

Do **not** allow the thermostat or thermometer to touch the container.

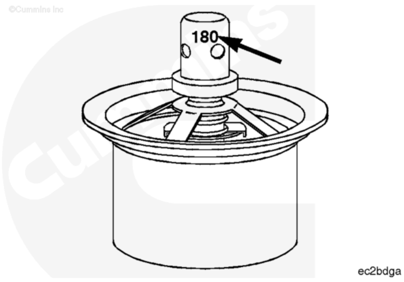
Heat the water and check the thermostat as follows.



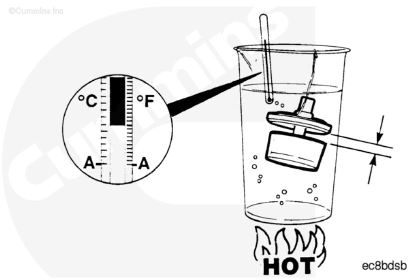
The nominal operating temperature is stamped on the thermostat.

- Thermostat **must** begin to open within 2°C [3°F] of nominal temperature.
- Thermostat **must** be fully open at 12°C [22°F] above nominal temperature.

The fully open distance between the thermostat flange and housing is 10.92 mm [0.43 in].



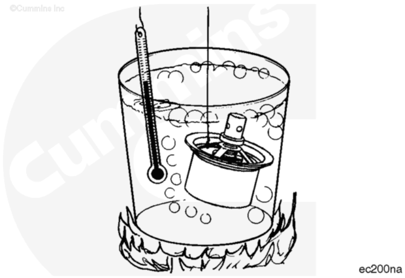
Remove the container from the heat. Check to see if the thermostat returns to the closed position.



Marine Applications

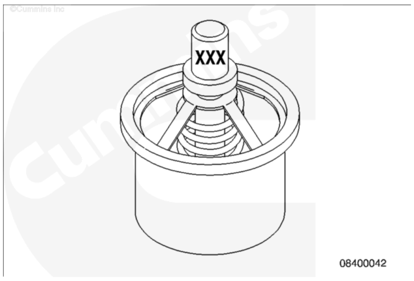
Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

Do **not** allow the thermostat or thermometer to touch the container.



Heat the water and check the thermostat as follows:

Nominal temperature is 66°C [150°F].



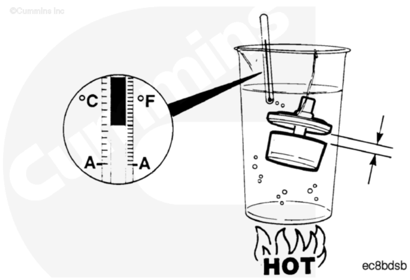
Measurements		
	celsius	fahrenheit
Thermostat begins to open:	63 to 67	145 to 152

Measurements		
	celsius	fahrenheit
Thermostat fully open:	79	175

The fully open distance is 9.5 mm [0.375 in].

Remove the container from heat.

Make sure the thermostat returns to the closed position.

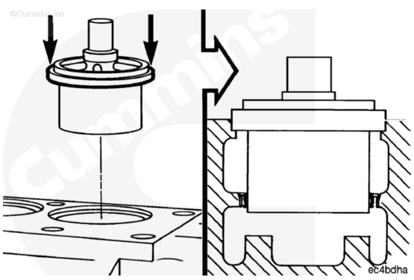


Install

Industrial Applications

Install the thermostat seals. Refer to Procedure 008-016 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-016.html)

Install the thermostat by pushing on the outer rim.

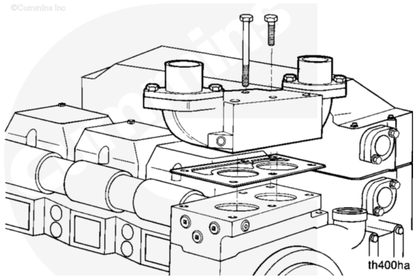


Install a new gasket.

Install the thermostat housing and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Use vegetable oil to lubricate the o-ring on the bypass tube.

Install the bypass tube.

Install the seal (10).

Install the retainer (9) and capscrew.

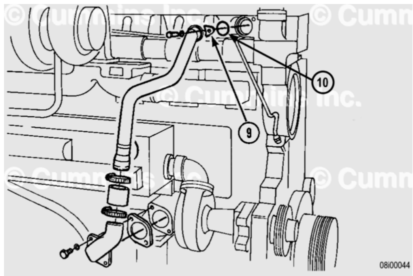
Tighten the capscrew and the hose clamps.

Torque Value:
Capscrew

1. 45 n•m [35 ft-lb]

Torque Value:
Hose Clamps

1. 6 n•m [50 in-lb]



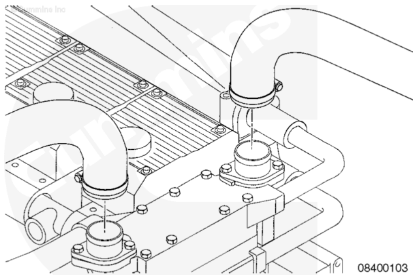
Install the two upper radiator hoses.

Tighten the hoses clamps.

Torque Value: 6 n•m [50 in-lb]

Install the vent lines.

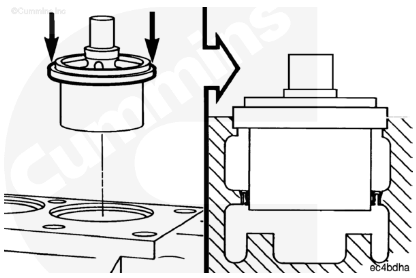
If the engine is equipped with an air compressor, install the air compressor coolant return tube.



Marine Applications

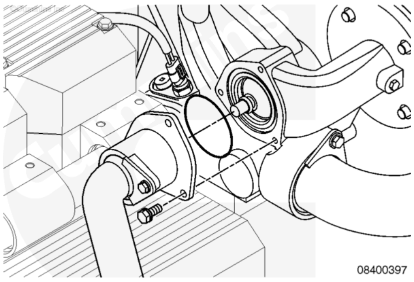
Install the thermostat seal. Refer to Procedure 008-016 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-016.html)

Install the thermostat by pushing on the outer rim until it seats.



Use a new o-ring and fasten the thermostat housing supply pipe to the thermostat housing with three captive washer capscrews.

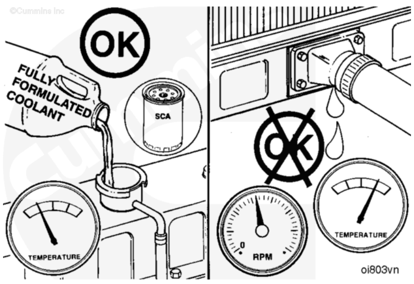
Torque Value: 40 n•m [30 ft-lb]



Finishing Steps

All Applications

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html)
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.



Last Modified: 25-Aug-2021